

The selected GNS representation of a pure-state class

`MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

def

A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class.

Statement

Let A be a unital complex C^* -algebra, with the positive order used for its states, and fix a pure state ω . A pure state is an extreme point of the normalized positive continuous linear functionals. Let I be the set of pure-state GNS representations modulo unitary equivalence. For each $j \in I$, choose a pure state φ_j in that class, retaining $\varphi_o = \omega$ literally at $o = [\omega]$. Write (H_j, π_j, ξ_j) for its GNS space, representation and canonical cyclic vector. The selected representation is the GNS action $\pi_j : A \rightarrow B(H_j)$ associated to this particular φ_j . The choice concerns the representative state; the construction then uses its actual GNS Hilbert space and action.

Definition

Take the chosen state φ_j , regard it as a positive linear functional, and form its GNS completion H_j . Left multiplication induces the bounded action π_j on that completion. The selected vector ξ_j is the GNS cyclic vector. Its orbit $\pi_j(A)\xi_j$ is dense. The vector-functional identity and purity of φ_j , together with this cyclicity and normalization, supply the separate irreducibility theorem.

Assumptions

The class $j \in I$ and the root pure state ω are given. GNS representations here are unital complex-linear star homomorphisms on complete complex Hilbert spaces. No separability or faithfulness of π_j is assumed.

Conclusion

The definition specifies π_j on H_j . The separate selected-vector and pure-GNS results give $\|\xi_j\| = 1$, $\varphi_j(a) = \langle \xi_j, \pi_j(a)\xi_j \rangle$, and irreducibility of π_j . These properties explain why this selected family supplies the summands of the direct-sum representation.

The inner product is conjugate linear in its first argument. Irreducibility means that the only closed reducing subspaces are 0 and H_j ; this Hilbert space is nonzero because it contains the unit vector ξ_j . The definition does not itself prove those properties, and it does not claim that different pure states always have inequivalent GNS representations.

Main citations

- Definition and its exact construction —
MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation
- Root-preserving choice of a pure state in each GNS class —
MathlibAnnex.CStarAlgebra.PureState.representative
- The selected GNS Hilbert space — MathlibAnnex.CStarAlgebra.PureState.SelectedGNS
- The selected cyclic vector — MathlibAnnex.CStarAlgebra.PureState.selectedVector
- Normalization of the selected cyclic vector —
MathlibAnnex.CStarAlgebra.PureState.norm_selectedVector
- The selected state as a vector functional —
MathlibAnnex.CStarAlgebra.PureState.selected_vectorFunctional
- Density of the selected GNS orbit —
MathlibAnnex.CStarAlgebra.PureState.denseRange_representative_gns_orbit
- Irreducibility supplied by purity and cyclicity —
MathlibAnnex.CStarAlgebra.PureState.isIrreducible_selectedRepresentation

Supporting route explanation

Lean source signature (exact)

```
noncomputable def selectedRepresentation (root : PureState A) (j : GNSClass A) :  
  MathlibAnnex.Analysis.CStarAlgebra.Representation A (SelectedGNS root j) :=  
  (representative root j).positiveFunctional.gnsStarAlgHom
```

In the source	Mathematical meaning
A; root : PureState A	The given unital complex C*-algebra A with its state order, and a fixed pure state ω . A pure state is a positive normalized continuous complex-linear functional extreme in the real convex state space.
j : GNSClass A	A class j of pure-state GNS representations modulo unitary equivalence; this class is not itself a state functional.
representative root j	The selected pure state φ_j belonging to class j , chosen with $\varphi_o = \omega$ at $o = [\omega]$.
SelectedGNS root j	The actual complete GNS Hilbert space H_j of that selected state φ_j .
Representation A (SelectedGNS root j)	The output is a unital complex-linear star homomorphism $\pi_j : A \rightarrow B(H_j)$.
(representative root j).positiveFunctional.gnsStarAlgHom	The full RHS takes the positive functional of φ_j and its GNS left-multiplication action on H_j . The vector ξ_j and the cyclicity, normalization and irreducibility properties have their separate cited definitions and theorems.

Further source notes: The vector ξ_j is named `selectedVector root j` in its separate cited definition; the normalization, vector-functional identity and irreducibility are separate cited theorems, not extra fields of this definition.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

Accepted content SHA-256: `03a298e788ee0db85d5beb7267b35048408970cddf4e2bccc8c0f5163ee51b84`

Accepted source guide SHA-256: `ce6f62005374b949f2186d0a6ac7e0ed796f2b9cbdb22c8fc03e4289ac0bdef7`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`